Random binning encoding and secrecy capacity

Laboratory session 2

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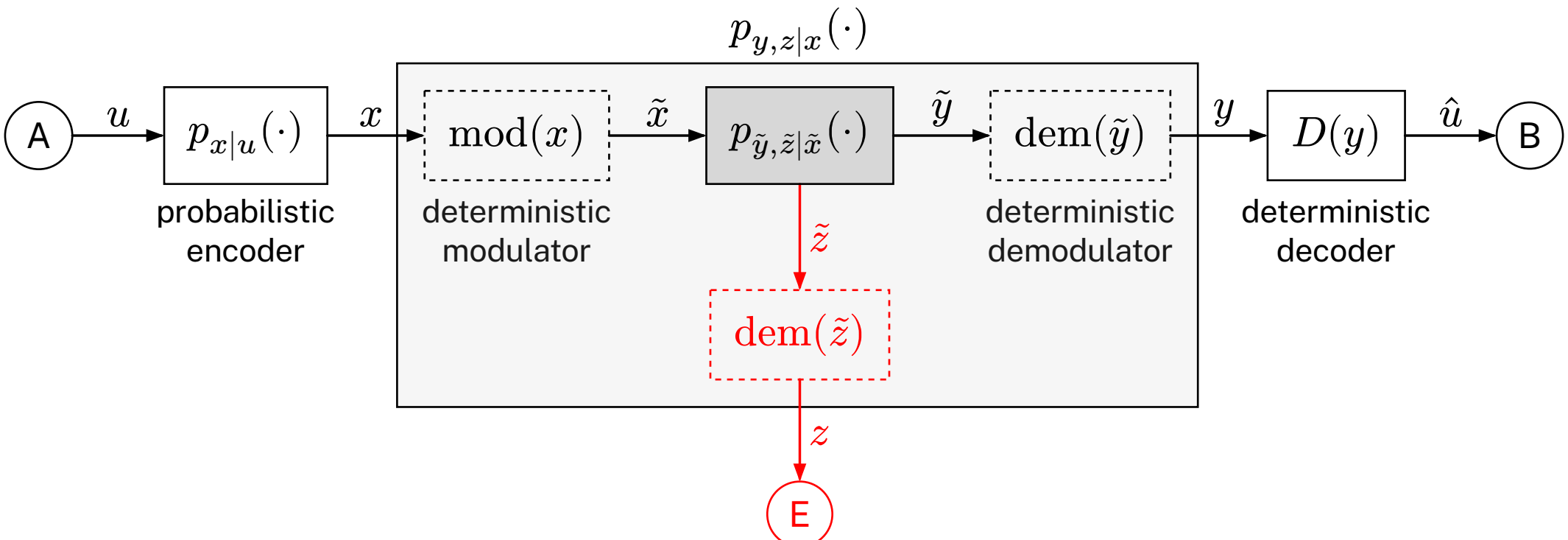
The wiretap channel



The wiretap uniform error channel

Definition (wiretap uniform error channel)

The wiretap uniform error channel is defined as follows:

- the legitimate channel introduces at most r binary errors per word,

$$T_{y|x}(a) = \{b \in \mathcal{Y} : d_{\mathsf{H}}(a, b) \leq r\};$$

- the eavesdropper channel introduces at most s binary errors per word,

$$T_{z|x}(a) = \{c \in \mathcal{Z} : d_{\mathsf{H}}(a, c) \leq s\};$$

- y and z are **conditionally uniform** and independent of each other given x.

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The wiretap binary symmetric channel

Definition (wiretap binary symmetric channel)

The memoryless wiretap binary symmetric channel is defined as follows:

- the legitimate channel has crossover probability ε per bit, i.e.
  $$p_{y|x}(\mathtt{0}|\mathtt{1}) = p_{y|x}(\mathtt{1}|\mathtt{0}) = \varepsilon, \quad p_{y|x}(\mathtt{0}|\mathtt{0}) = p_{y|x}(\mathtt{1}|\mathtt{1}) = 1 - \varepsilon;$$
- the eavesdropper channel has crossover probability δ per bit, i.e.
  $$p_{z|x}(\mathtt{0}|\mathtt{1}) = p_{z|x}(\mathtt{1}|\mathtt{0}) = \delta, \quad p_{z|x}(\mathtt{0}|\mathtt{0}) = p_{z|x}(\mathtt{1}|\mathtt{1}) = 1 - \delta.$$

Theorem (secrecy capacity for a wiretap BSC)

A memoryless wiretap binary symmetric channel has secrecy capacity

$$C_{\mathsf{s}} = \begin{cases} 0 & \text{if } |\varepsilon - \frac{1}{2}| \leq |\delta - \frac{1}{2}| \\ h_2(\delta) - h_2(\varepsilon) & \text{if } |\varepsilon - \frac{1}{2}| > |\delta - \frac{1}{2}|, \end{cases}$$

where $h_2(\cdot)$ is the binary entropy function.

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The wiretap AWGN channel

Definition (wiretap AWGN channel)

The memoryless wiretap AWGN channel is defined as follows:

- the legitimate channel has signal-to-noise ratio $\mathrm{SNR}_{\mathsf{B}}$ (in dB), i.e.
$$y = x + \mathcal{N}\big(0, P \cdot 10^{-\mathrm{SNR}_{\mathsf{B}}/10}\big);$$
- the eavesdropper channel has signal-to-noise ratio $\mathrm{SNR}_{\mathsf{E}}$ (in dB), i.e.
$$z = x + \mathcal{N}\big(0, P \cdot 10^{-\mathrm{SNR}_{\mathsf{E}}/10}\big).$$

Theorem (secrecy capacity for a wiretap AWGN channel)

A memoryless wiretap AWGN channel has secrecy capacity

$$C_{\mathsf{s}} = \begin{cases} 0 & \text{if } \mathrm{SNR}_{\mathsf{B}} \leq \mathrm{SNR}_{\mathsf{E}} \\ \frac{1}{2}\log_2\left(\frac{1+10^{\mathrm{SNR}_{\mathsf{B}}/10}}{1+10^{\mathrm{SNR}_{\mathsf{E}}/10}}\right) & \text{if } \mathrm{SNR}_{\mathsf{B}} > \mathrm{SNR}_{\mathsf{E}}. \end{cases}$$

Task 1 — Implement the wiretap uniform error channel

Task 1

Using a programming language of your choice, implement a wiretap uniform error channel according to the following parameters.

- Input and output alphabets: $\mathcal{X} = \mathcal{Y} = \mathcal{Z} = \{\texttt{0}, \texttt{1}\}^7$
- Legitimate channel binary error per word upper bound: $r = 1$
- Eavesdropper channel binary error per word upper bound: $s = 3$

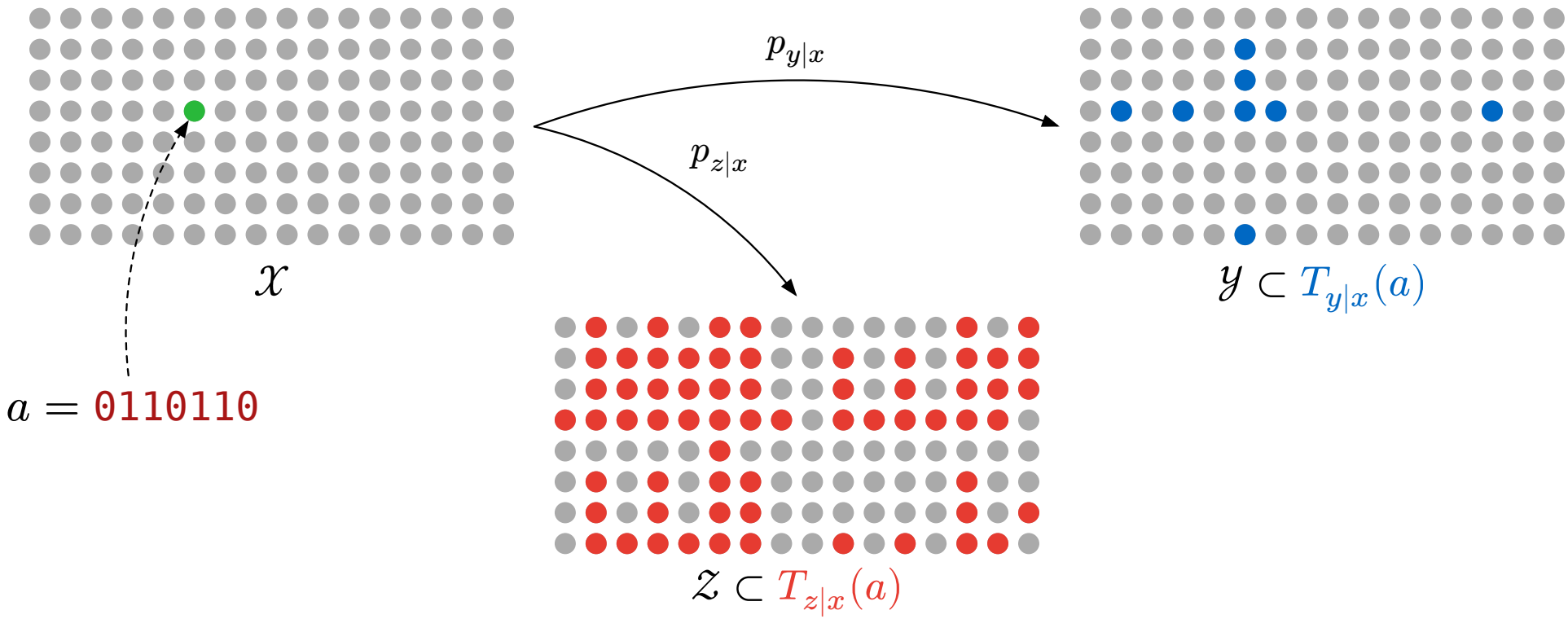
Simulate a sufficiently large number (at least 2^{14}) of channel realizations with the same input $x = \texttt{1001000}$, then:

- plot the empirical distributions $\hat{p}_{y|x}(\cdot \mid \texttt{1001000})$ and $\hat{p}_{z|x}(\cdot \mid \texttt{1001000})$;
- verify the conditional independence of y and z numerically.

Task 1: interpretation

$$y = x \oplus \mathcal{U}(\{\texttt{0000000}, \texttt{0000001}, \texttt{0000010}, \ldots, \texttt{1000000}\})$$

$$z = x \oplus \mathcal{U}(\{\texttt{0000000}, \texttt{0000001}, \ldots, \texttt{0000011}, \ldots, \texttt{1101000}, \texttt{1110000}\})$$



Task 2 — Implement the random binning encoder

Task 2

Using a programming language of your choice, implement a random binning encoder according to the following parameters.

- Message space: $\mathcal{M} = \{\texttt{0}, \texttt{1}\}^3$
- Encoder output space: $[7, 4, 3]$ Hamming code $\mathcal{X}' \subset \mathcal{X}$, i.e.
  $\mathcal{X}' = \{\texttt{0000000}, \texttt{1000110}, \texttt{0100101}, \texttt{1100011}, \texttt{0010011}, \texttt{1010101}, \texttt{0110110}, \texttt{1110000},$
  $\texttt{0001111}, \texttt{1001001}, \texttt{0101010}, \texttt{1101100}, \texttt{0011100}, \texttt{1011010}, \texttt{0111001}, \texttt{1111111}\}$
- Encoding rule: x is chosen uniformly at random from the 2-codeword bin
  $T_{x|u}(d) = \left\{a \in \mathcal{X}' : a \text{ prefix is either } \texttt{0}\|d \text{ or } \texttt{1}\|\bar{d}\right\}$ where $\bar{d} = (d \oplus \texttt{111})$

Verify the correctness of your implementation by observing the codewords produced for several different messages.

Task 3 — Implement the random binning decoder

Task 3

Using a programming language of your choice, implement a random binning deterministic decoder $D : \mathcal{Y} \to \mathcal{M}$ for the encoder from Task 2. In particular, the decoder should:

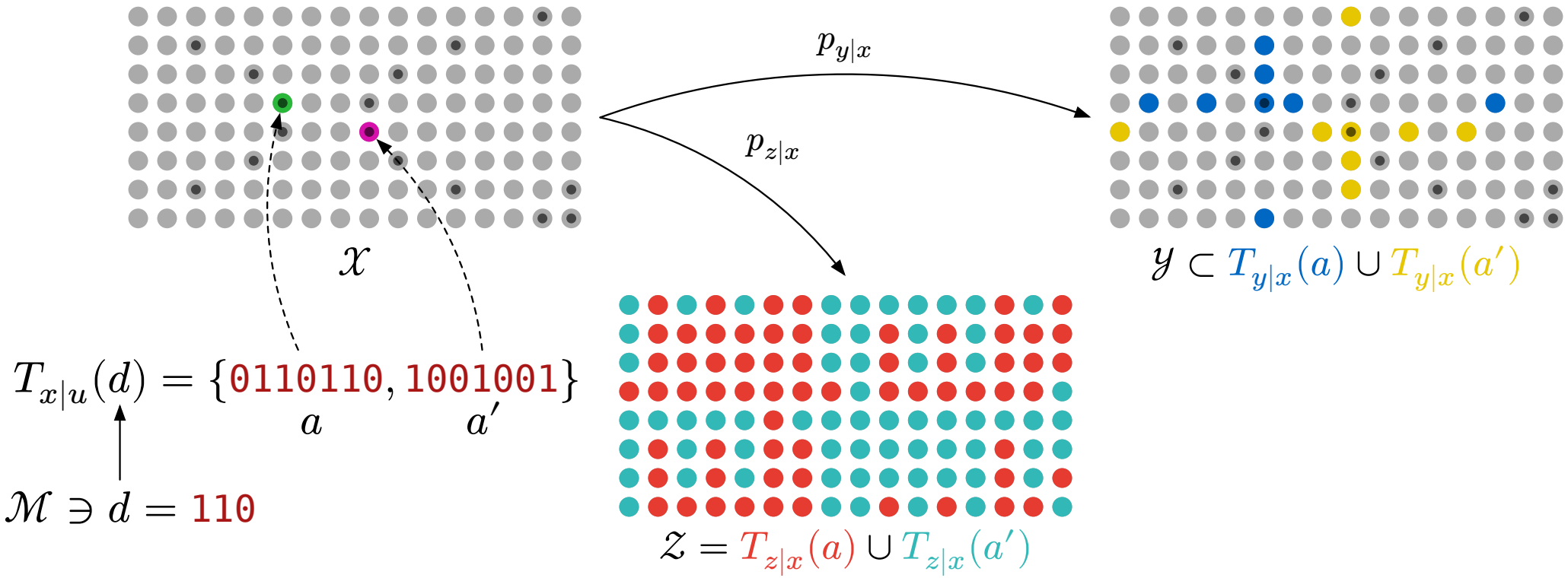
1. identify the transmitted codeword by the minimum Hamming distance criterion
$$\hat{x} = \arg\min_{a \in \mathcal{X}'} d_{\mathsf{H}}(y, a);$$
2. look at the first bit $\hat{x}_1$ of $\hat{x}$ and identify the transmitted message $\hat{u}$ as either the bits $\hat{x}_2 \| \hat{x}_3 \| \hat{x}_4$ in $\hat{x}$ or their ones' complement.

Verify the correctness of your implementation by:

- cascading encoder + decoder;
- cascading encoder + legitimate channel + decoder.

} no errors expected

Tasks 2–3: interpretation



Task 4 — Show perfect secrecy

Task 4

Simulate a sufficiently large number (at least 2^{14}) of realizations of the encoder + eavesdropper channel chain, then:

- plot the probability mass function $p_{z|u}(\cdot\,|d)$ for all values of $d \in \mathcal{M}$;
- compute the empirical joint distribution $\hat{p}_{u,z}(d,c)$;
- compute the empirical marginal distributions $\hat{p}_u(d)$ and $\hat{p}_z(c)$;
- compute the empirical entropy $H(u)$;
- compute the empirical mutual information

$$\hat{I}(u;z) = \sum_{d\in\mathcal{M},c\in\mathcal{Z}} \hat{p}_{u,z}(d,c)\log_2 \frac{\hat{p}_{u,z}(d,c)}{\hat{p}_u(d)\hat{p}_z(c)}.$$

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Task 5 — Discuss the properties of the system

Task 5

Discuss the properties of the system you realized in Tasks 1–4, using the following questions as guidelines.

1. Can you say that your system provides perfect secrecy?
2. How many secret message bits per channel use (i.e. per transmitted word) have you obtained with your scheme?
3. Is it possible to obtain more secret bits per channel use?
   If so, how should you change your encoder/decoder? If not, why?
4. One could consider evaluating the secrecy of this mechanism by cascading encoder + eavesdropper channel + decoder and measuring the error rate. Why should one evaluate the (more complicated) mutual information instead?

Task 6 — Implement the wiretap binary symmetric channel

Task 6

Using a programming language of your choice, implement a wiretap binary symmetric channel according to the following parameters.

- Input and output alphabets: $\mathcal{X} = \mathcal{Y} = \mathcal{Z} = \{\mathtt{0}, \mathtt{1}\}^7$
- Legitimate channel binary error probability: arbitrary ε set upon instantiation
- Eavesdropper channel binary error probability: arbitrary δ set upon instantiation

Verify the correctness of your implementation by transmitting a long binary sequence and measuring the average number of bit errors in each output, for several values of ε and δ.

Task 7 — Evaluate the system security over the wiretap BSC

Task 7

Choose several values of ε and δ, and for each (ε, δ) pair simulate a sufficiently large number (at least 2^{14}) of realizations of the appropriate chains, then:

- evaluate the system reliability by plotting the error decoding probability $\mathrm{P}[\hat{u} \neq u]$ as a function of ε;
- evaluate the system secrecy by plotting the empirical leaked information $\hat{I}(u; z)$ as a function of δ;
- using the results from the previous points, compute and plot an upper bound on the unconditional security of the system;
- compute and plot the empirical total variation distance between your system and the ideal counterpart $d_{\mathrm{V}}(p_{u\hat{u}z}, p_{u\hat{u}^\star z^\star})$, and compare the result with the previous point;
- measure and plot the secrecy capacity of the system.

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Task 8 — Implement the wiretap AWGN channel

Task 8

Using a programming language of your choice, implement an AWGN wiretap channel, employing a PAM-2^{ℓ_x} modulation, according to the following parameters.

- Input and output alphabets: $\mathcal{X} = \mathcal{Y} = \mathcal{Z} = \{\texttt{0}, \texttt{1}\}^7$
- Legitimate channel SNR: arbitrary $\mathrm{SNR}_{\mathsf{B}}$ set upon instantiation
- Eavesdropper channel SNR: arbitrary $\mathrm{SNR}_{\mathsf{E}}$ set upon instantiation
- PAM modulator output and demodulator input alphabets: $\tilde{\mathcal{X}} = \tilde{\mathcal{Y}} = \tilde{\mathcal{Z}} = \mathbb{R}$
- PAM cardinality: $M = 2^{\ell_x} = 2^7$

Verify the correctness of your implementation by transmitting a long symbol sequence and comparing the ratio of the average number of symbol errors in the two outputs to the square root of the ratio of the two linear SNRs, for several values of $\mathrm{SNR}_{\mathsf{B}}$ and $\mathrm{SNR}_{\mathsf{E}}$.

Task 9 — Evaluate the system security over the wiretap AWGN channel

Task 9

Choose several values of $\mathrm{SNR}_{\mathsf{B}}$ and $\mathrm{SNR}_{\mathsf{E}}$, and for each $(\mathrm{SNR}_{\mathsf{B}}, \mathrm{SNR}_{\mathsf{E}})$ pair simulate a sufficiently large number (at least 2^{14}) of realizations of the appropriate chains, then:

- evaluate the system reliability by plotting the error decoding probability $\mathrm{P}[\hat{u} \neq u]$ as a function of ε;
- evaluate the system secrecy by plotting the empirical leaked information $\hat{I}(u; z)$ as a function of δ;
- using the results from the previous points, compute and plot an upper bound on the unconditional security of the system;
- compute and plot the empirical total variation distance between your system and the ideal counterpart $d_{\mathsf{V}}(p_{u\hat{u}z}, p_{u\hat{u}^\star z^\star})$, and compare the result with the previous point;
- measure and plot the secrecy capacity of the system.

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What you need to submit

Each group must turn in, through the Moodle assignment submission procedure:

1. the code of your implementations (well organized, as a compressed archive file);
2. a short report in PDF format, including:
   - 2.1. a brief description of your implementations, explaining your choices and any challenges your encountered;
   - 2.2. plots and numerical results, accompanied by detailed qualitative interpretations;
   - 2.3. your discussion for Task 5 including precise answers to the guideline questions.

Use of artificial intelligence tools must be adequately justified and disclosed with proper attribution. Group members are fully responsible for the submitted work.